Section 4 — Scenario-Based Testing (Q51–Q75)

Test Scenarios, Test Cases & Edge Cases

51. How would you test a login page?

Test Scenarios

- Verify the login page UI (all elements, labels, alignment)
- Verify login with valid credentials (positive)
- Verify login with invalid credentials (negative)
- Verify field-level validations (empty / invalid format)
- Verify account lock after repeated failures
- Verify session behaviour after login and logout
- Verify 'Remember Me' functionality
- Verify security aspects (masked password, SQL injection, back button)

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
51.01	UI verification	Verify all elements load: username field, password field, Login button, Forgot Password link, Remember Me checkbox	All elements visible with correct labels and alignment
51.02	Positive	Login with valid username and password	User logs in successfully and lands on dashboard/home page
51.03	Negative	Login with valid username and wrong password	Error shown: 'Invalid username or password'; login blocked
51.04	Negative	Login with unregistered username	Same generic error; no hint that username does not exist
51.05	Field validation	Submit with empty username, password filled	Validation message: 'Username is required'
51.06	Field validation	Submit with both fields empty	Both field-level validation messages shown; no request sent
51.07	Negative	Password entered in wrong case (e.g., 'PASSword' vs 'Password')	Login fails; password is case-sensitive
51.08	UI / security	Password field display	Password masked as dots/asterisks; Show/Hide toggle works
51.09	Account lock	Enter wrong password 3-5 times consecutively	Account locked or CAPTCHA shown as per policy
51.10	Session	Press browser Back button after logging out	User not taken back into logged-in page; redirected to login
51.11	Security	Enter ' OR '1'='1 in username/password	Login fails; no SQL injection possible
51.12	Remember Me	Login with Remember Me checked, close and reopen browser	User stays logged in for the remember period
51.13	Usability	Use Tab key to move between fields; press Enter to submit	Tab order is logical; Enter submits the form
51.14	Compatibility	Login on Chrome, Firefox, Safari, Edge and mobile browser	Works and displays correctly on all browsers

Edge Cases

Edge ID	Edge Case	Expected Behaviour
51-E1	Login with email containing leading/trailing spaces	Trimmed or rejected consistently; same result every time

Edge ID	Edge Case	Expected Behaviour
51-E2	Unicode characters / emojis typed in username	Handled gracefully; no crash or broken error message
51-E3	Password field filled to 300+ characters (paste)	Blocked at max length or rejected by server with clear message
51-E4	Login from two devices at the same moment (concurrent)	Both handled per policy; no race condition or data corruption
51-E5	Caps Lock ON while typing password	Ideally warned by UI; login fails cleanly if ignored
51-E6	Submit login exactly as session/CSRF token expires	Clean re-authentication prompt, not an unhandled error
51-E7	Browser autofill injects stale saved credentials	Autofilled values validated the same as typed values

52. Write test scenarios for a login page.

Test Scenarios

- Functional scenarios: valid login, invalid username, invalid password, empty fields
- UI scenarios: layout, labels, tab order, Enter key, show/hide password
- Security scenarios: masking, SQL injection, session timeout, back-button after logout
- Account-lock scenario after max failed attempts
- Remember Me and Forgot Password link scenarios
- Compatibility and performance scenarios (different browsers, load)

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
52.01	Functional	Login with valid credentials → verify successful navigation	Dashboard opens with user name displayed
52.02	Functional	Login with invalid username / invalid password / both	Appropriate error message; no entry to system
52.03	UI	Verify placeholder text, tab order, Enter key, password masking	Each UI element behaves as specified
52.04	Security	Session timeout after inactivity (e.g., 15 min)	User auto-logged out; must log in again
52.05	Security	Try common injections and script tags in fields	Input rejected/sanitized; no error page exposed
52.06	Lockout	5 consecutive failed logins	Account locked, message shown, unlock flow available
52.07	Links	Click 'Forgot Password'	Forgot password page opens correctly
52.08	Compatibility	Login across browsers and mobile view	Consistent behaviour everywhere

Edge Cases

Edge ID	Edge Case	Expected Behaviour
52-E1	Double-click the Login button rapidly	Only one authentication request; no duplicate session
52-E2	Login during scheduled server maintenance	Maintenance message shown, no raw error page
52-E3	Backspace editing inside the masked password field	Edits correctly; characters removed, not overwritten
52-E4	Login page opened from a very old cached bookmark/URL	Page loads fresh; stale tokens are not honoured
52-E5	Very slow network while submitting (2G throttle)	Spinner shown; no double submit; times out with friendly message

53. What negative test cases would you create for login?

Test Scenarios

- Wrong credentials: invalid password, unregistered username, both wrong
- Empty fields: username only, password only, both
- Malformed input: spaces, invalid format, very long input, special characters
- Lockout behaviour: repeated failures, attempt counter
- Security negatives: SQL injection, expired/locked account, copy-paste tricks

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
53.01	Wrong credentials	Valid username + wrong password	Login fails with generic error message
53.02	Wrong credentials	Unregistered username + any password	Login fails; error does not reveal which field is wrong
53.03	Empty fields	Leave username blank, fill password	'Username required' validation; login blocked
53.04	Empty fields	Leave password blank	'Password required' validation; login blocked
53.05	Empty fields	Both fields blank and click Login	Both validations fire; no server call
53.06	Malformed input	Username with leading/trailing spaces	Handled per spec (trimmed or rejected) consistently
53.07	Malformed input	Paste 200+ characters into username field	Rejected or truncated per field limit
53.08	Malformed input	Special characters / script tags in username	Rejected gracefully; no crash or error page leak
53.09	Lockout	Wrong password 5 times in a row	Account locked or CAPTCHA triggered; counter shown if specified
53.10	Account state	Login with locked / deactivated account	Clear message: 'Account locked. Contact support'
53.11	Security	Enter SQL injection strings like ' OR '1'='1	Login rejected; input sanitized
53.12	Security	Use URL of the inner page directly without logging in	Redirected to login page (session enforced)

Edge Cases

Edge ID	Edge Case	Expected Behaviour
53-E1	Whitespace-only username (" ")	Treated as empty or rejected — consistent behaviour
53-E2	NULL byte or control characters in input	Rejected/sanitized; no server error leaked
53-E3	Credentials of a just-deleted account	Generic 'invalid' response; no account-status leak
53-E4	SQL wildcards (% _) in username	Treated as literal text; no query behaviour change
53-E5	Failed-attempt counter exactly at limit minus one, then correct password	Login succeeds; counter resets correctly

54. How would you test Forgot Password?

Test Scenarios

- Request reset link with a registered email
- Request with unregistered / invalid / empty email
- Reset the password using a valid link and set a new one
- Link expiry and single-use behaviour
- New password rules and re-login with the new password

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
54.01	Positive	Enter registered email and submit	Reset link sent message shown; email received with working link
54.02	Negative	Enter unregistered email	Generic message: 'If the email exists, a link has been sent' (no user enumeration)
54.03	Validation	Enter invalid email format (abc@)	Inline validation error shown
54.04	Validation	Submit with empty email	'Email is required' message
54.05	Positive	Open link, enter valid new password + matching confirm	Password updated; confirmation message shown
54.06	Negative	New password and confirm password do not match	Error shown; password not changed
54.07	Negative	New password that violates rules (too short, no special char)	Rule-specific error message displayed
54.08	Link expiry	Use the reset link after expiry time (e.g., 30 min)	Link expired message; user must request a new one
54.09	Single-use	Use the same reset link a second time	Link already used / invalid; new request required
54.10	Old link invalid	Use an older link after requesting a new one	Old link is invalid; only the latest works
54.11	Re-login	Login with the old password after reset	Login fails; old password invalidated
54.12	Re-login	Login with the new password	Login succeeds
54.13	Security	Change reset token value manually in URL	Link rejected as invalid
54.14	Rate limit	Request reset link many times rapidly	Rate limiting / cooldown applied as per spec

Edge Cases

Edge ID	Edge Case	Expected Behaviour
54-E1	Reset requested from email alias (user+tag@gmail.com)	Alias matched to same account as per spec
54-E2	Two reset links requested; both opened at once in tabs	Only latest link valid; older one rejected
54-E3	Reset link opened exactly at the expiry moment	Treated as expired — no ambiguity
54-E4	Password changed on device A while reset done on device B	Both flows conflict safely; latest change wins, no corruption
54-E5	Email service delayed — user requests again before first arrives	Second request honoured; both emails usable only per single-use rule
54-E6	Reset link opened on a different browser/device than requested	Either allowed or flagged per security policy — consistently enforced

55. How would you test user registration?

Test Scenarios

- Successful registration with all valid data
- Mandatory-field validation (each required field left empty)
- Duplicate user (same email / username) prevention
- Password and confirm-password rule enforcement
- Field format and length validations (email, mobile, name)
- Terms checkbox and post-registration actions (welcome email, login)

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
55.01	Positive	Fill all fields with valid data, accept terms, submit	Account created; redirected to login/dashboard
55.02	Mandatory	Submit with each mandatory field empty one at a time	Field-specific 'required' message each time
55.03	Duplicate	Register with an already-registered email	'Email already exists' error; no second account
55.04	Duplicate	Register with existing username	'Username taken' error with suggestion if specified
55.05	Password rules	Enter weak password (only letters, 4 chars)	All policy violations listed clearly
55.06	Password match	Confirm password differs from password	'Passwords do not match' error
55.07	Format	Enter invalid email / mobile format	Inline format validation errors
55.08	Length	Exceed max length in name (e.g., 300 chars)	Rejected or blocked at field limit with message
55.09	Terms	Submit without checking the Terms checkbox	Registration blocked with message
55.10	Post-reg	Check welcome/verification email after signup	Email received; verification link works
55.11	Data check	Verify saved data in database/profile after registration	Stored values match exactly what was entered
55.12	Special chars	Enter special characters / emojis in name field	Accepted or rejected consistently per spec

Edge Cases

Edge ID	Edge Case	Expected Behaviour
55-E1	Two users register with same email at the same instant	One succeeds, one gets 'already exists'; no duplicate accounts
55-E2	Birth date 29/02 (leap day) if DOB field exists	Accepted; validates correctly on leap/non-leap years
55-E3	Name field containing only spaces	Rejected as empty or sanitized per spec
55-E4	Registration while email/SMS service is down	Account created per policy; verification retried or queued
55-E5	Password exactly at min and max allowed length (boundary)	Both accepted; min−1 and max+1 rejected
55-E6	Age boundary if applicable (exactly 18 or exactly 60)	Boundary values accepted/rejected exactly per rule

56. How would you test an OTP feature?

Test Scenarios

- Correct OTP entered within validity
- Incorrect OTP and max-attempt limits
- OTP expiry (time-based)
- Resend OTP (cooldown, new OTP invalidates old)
- OTP delivery channel and content correctness

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
56.01	Positive	Enter correct OTP within validity time	Verification succeeds; user proceeds
56.02	Negative	Enter wrong OTP	Error: 'Invalid OTP'; attempt count shown if specified
56.03	Lockout	Enter wrong OTP up to max attempts (e.g., 3)	Blocked / re-request flow after limit; correct OTP then also rejected
56.04	Expiry	Enter correct OTP after expiry (e.g., 60s+ later)	'OTP expired' error; user must request new OTP
56.05	Resend	Click Resend immediately after first OTP	Cooldown enforced (e.g., wait 30s) or new OTP sent per spec
56.06	Resend	Request new OTP, then enter the OLD OTP	Old OTP is invalid; only newest works
56.07	Input handling	Enter OTP with spaces or lowercase (if alphanumeric)	Handled per spec — trimmed / case-insensitive as designed
56.08	Delivery	Verify OTP arrives on registered mobile/email	OTP delivered on the correct channel with correct template
56.09	Auto-read/UI	OTP auto-fill on Android / paste into 4-6 boxes	Boxes accept digits; auto-advance between boxes
56.10	Expiry timer	Watch countdown timer in UI	Timer matches validity period; submit disabled when expired
56.11	Blank/invalid	Submit empty OTP or letters where digits expected	Validation blocks submission

Edge Cases

Edge ID	Edge Case	Expected Behaviour
56-E1	OTP submitted at the exact expiry second	Single deterministic outcome — accepted or expired, never random
56-E2	Resend clicked exactly when cooldown ends	New OTP sent; no double-send or blocked-state deadlock
56-E3	SMS arrives late, after OTP already expired	Expired OTP rejected with clear 'expired' message
56-E4	O vs 0 confusion in alphanumeric OTP (copy-paste)	Exact-match enforcement; clear error for wrong value
56-E5	Resend requested from two tabs simultaneously	One OTP generated; no duplicate messages/valid codes
56-E6	Device time zone changed between request and entry	Expiry based on server time; no bypass via device clock

57. How would you test an e-commerce Add to Cart feature?

Test Scenarios

- Add a single in-stock product to cart
- Add multiple different products / multiple quantities
- Add the same product again (quantity increment vs duplicate row)
- Add out-of-stock / unavailable products
- Cart badge and price update after add
- Add to cart as guest vs logged-in user
- Product variants (size/colour) selection before adding

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
57.01	Positive	Add one in-stock product to cart	Product appears in cart with correct image, name, price
57.02	Positive	Add 3 different products	All 3 shown in cart with correct details
57.03	Quantity	Add same product twice	Quantity becomes 2 in a single row (no duplicate line item)
57.04	Quantity	Set quantity to 0 or negative before adding	Blocked with validation message
57.05	Stock	Try to add a product with 0 stock	'Out of stock' shown; cannot add
57.06	Stock	Add quantity greater than available stock	Capped at available stock or error shown
57.07	UI	Check cart badge/counter after each add	Badge count = total quantity in cart
57.08	Variant	Add same product in different size/colour	Separate line items per variant
57.09	Guest	Add to cart without logging in	Product added; cart merges/persists after login as per spec
57.10	Price	Add product on discount	Correct discounted price shown in cart
57.11	Navigation	Add to cart and continue shopping	Cart retains item while browsing
57.12	Speed	Click Add to Cart twice quickly	No duplicate charge/row beyond qty+1; no UI glitch

Edge Cases

Edge ID	Edge Case	Expected Behaviour
57-E1	Add to cart at the exact moment stock hits 0 (race)	Either added or blocked cleanly; cart and stock never go negative
57-E2	Quantity set to 999+ on a low-stock product	Capped at available stock with clear message
57-E3	Price changes (sale starts) between page load and add	Cart shows the price at time of add per business rule
57-E4	Same cart open in two tabs; add from both	Cart merges correctly; no lost or duplicated lines
57-E5	Product deleted by admin while on its page	Graceful 'no longer available' message; no broken add

58. How would you test a shopping cart?

Test Scenarios

- Cart contents display (items, price, quantity, images)
- Update quantity of an item
- Remove an item / empty the cart
- Cart total and subtotal calculation
- Cart persistence (login, session, refresh, cross-device)
- Coupon/tax/shipping application on cart totals
- Out-of-stock / price-changed items in cart

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
58.01	Display	Open cart with items	Each item shows correct image, name, unit price, qty, line total
58.02	Quantity	Increase quantity from 1 to 2	Line total and cart total update correctly
58.03	Quantity	Decrease quantity to 0 or below	Item removed or minimum quantity enforced
58.04	Remove	Click Remove on an item	Item removed; totals recalculated
58.05	Empty cart	Remove all items / open cart with no items	'Your cart is empty' message with shopping CTA
58.06	Total	Verify subtotal, tax, shipping and grand total maths	Totals match expected calculation for the data
58.07	Coupon	Apply valid coupon in cart	Discount deducted; total reduced by correct amount
58.08	Persistence	Refresh page / log out and back in	Cart items retained (as per spec for the user/guest)
58.09	Stock	An item in cart goes out of stock	Cart flags the item and blocks checkout for it
58.10	Price change	Item price changes after adding	Current or locked price shown per business rule
58.11	Save for later	Move item to wishlist / save for later	Item leaves cart, appears in saved list, can be moved back
58.12	Navigation	Proceed to checkout from cart	Checkout opens with the same items and totals

Edge Cases

Edge ID	Edge Case	Expected Behaviour
58-E1	Tax/discount rounding (e.g., 10% of ₹99.99)	Rounds per spec; totals always sum exactly from line items
58-E2	Same item in guest cart AND saved cart at login	Quantities merged into one line; no duplicate rows
58-E3	Coupon makes total exactly ₹0 or negative	Clamped at ₹0; checkout continues or blocked per policy
58-E4	Item quantity updated to exactly the last stock unit	Accepted; subsequent increase blocked
58-E5	Currency changed mid-cart (if multi-currency)	Converted with correct rate and message; no mixed totals

59. How would you test an online payment page?

Test Scenarios

- Successful payment with a valid card / UPI / netbanking
- Card validation failures (invalid number, expired, wrong CVV)
- Declined payment and insufficient funds
- Payment failure handling and refund
- Double submission / duplicate charge prevention
- Security and data handling (masking, no card data leak)

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
59.01	Positive	Pay with a valid test card with sufficient balance	Payment succeeds; success page + order confirmation
59.02	Validation	Enter invalid card number (fails Luhn check)	'Invalid card number' error
59.03	Validation	Enter expired card (past date)	'Card expired' error
59.04	Validation	Wrong CVV or missing fields	Field-level errors; payment not attempted
59.05	Declined	Use a declined test card	Failure message; user stays on payment options; retry allowed
59.06	Amount	Pay when account has insufficient funds	Declined with clear reason; no partial deduction
59.07	Duplicate	Double-click Pay button rapidly	Only one transaction created; no double charge
59.08	Failure handling	Simulate gateway timeout / network drop mid-payment	Order pending state resolved; auto-refund if money left account
59.09	Refund	Payment fails after money deducted	Refund initiated; amount returns within stated SLA
59.10	Methods	Verify UPI, netbanking, wallet options	Each method completes with its own flow correctly
59.11	Security	Card number masked on review and in confirmation	Full number never displayed or emailed
59.12	Session	Open payment page in two tabs, pay in one	Second tab invalidated; no double order
59.13	Receipt	Check confirmation email/SMS after payment	Correct amount, order id and items in receipt

Edge Cases

Edge ID	Edge Case	Expected Behaviour
59-E1	Card expiring at end of the current month	Accepted on last valid day; rejected from next month
59-E2	Network drops after auth but before capture	Resolved to success or auto-refund; never stuck 'pending' forever
59-E3	Amount with paise (₹1,000.50)	Charged exactly; no rounding errors in receipt
59-E4	Session expires exactly on the payment screen	Re-authentication flow preserves the pending order; no double pay
59-E5	Same order paid in two parallel tabs	One payment succeeds; second blocked — never double-charged

60. How would you test a bank transaction (fund transfer)?

Test Scenarios

- Successful transfer to a valid beneficiary
- Insufficient balance handling
- Invalid amounts and limits (zero, negative, above daily limit)
- Beneficiary validation (invalid account, same account)
- Receipt, history and balance accuracy after transfer
- Failure/interruption handling (network drop mid-transfer)

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
60.01	Positive	Transfer valid amount to a valid added beneficiary	Money moves; success message with transaction reference
60.02	Balance	Transfer more than available balance	Transaction declined with 'insufficient funds'
60.03	Invalid amount	Enter 0 or negative amount	Blocked with validation message
60.04	Limits	Transfer above per-transaction / daily limit	Rejected with limit message; remaining limit shown per spec
60.05	Beneficiary	Transfer to invalid / closed account	Rejected by system with appropriate error
60.06	Beneficiary	Transfer to your own account number	Allowed or blocked as per bank policy — consistent behaviour
60.07	OTP/2FA	Complete transfer without / with wrong OTP	Transfer blocked until correct second-factor
60.08	Accuracy	Verify balances after transfer	Sender debited, receiver credited by exactly the amount; charges as per policy
60.09	History	Check transaction history after transfer	Entry shows correct date, amount, type, beneficiary, status
60.10	Receipt	Download/email transaction receipt	Receipt has correct reference number and details
60.11	Interruption	Network drops after OTP confirm	No partial debit; either completes or fails cleanly with record
60.12	Concurrency	Same account transfers from two sessions simultaneously	No race condition; balances remain correct

Edge Cases

Edge ID	Edge Case	Expected Behaviour
60-E1	Transfer exactly equal to remaining balance (leaves ₹0)	Succeeds if charges are zero; otherwise declined with clear maths shown
60-E2	Transfer at exactly midnight / daily-limit reset time	Belongs to exactly one day's limit; no double or zero allowance
60-E3	Two transfers submitted in the same second	Both execute sequentially; limits and balance stay correct
60-E4	Beneficiary account closed between validation and debit	Transaction fails safely; money not lost (reversal if needed)
60-E5	Amount with paise (₹500.55)	Transferred exactly; no silent rounding

61. How would you test an ATM?

Test Scenarios

- Card insertion and PIN verification
- Wrong PIN / card blocked after 3 attempts
- Cash withdrawal within limits (per-transaction, daily, balance)
- Dispensing correctness (denominations, partial availability)
- Receipt, balance enquiry, and other services
- Hardware/exception behaviour (no cash, card retention, power cut, door)

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
61.01	Positive	Insert valid card, enter correct PIN, withdraw valid amount	Cash dispensed exactly; card returned; balance reduced correctly
61.02	Negative	Enter wrong PIN once	Error shown; allowed to retry
61.03	Lockout	Enter wrong PIN 3 times	Card blocked/retained as per policy; message shown
61.04	Limits	Withdraw above per-transaction limit (e.g., >10k or >25 notes worth)	Rejected; limit message with suggestion
61.05	Limits	Withdraw above daily limit	Rejected even if balance is sufficient
61.06	Balance	Withdraw more than account balance	'Insufficient funds' error; no cash dispensed
61.07	Denominations	Withdraw an amount the machine cannot make with available notes (e.g., 175)	Message asks to choose multiples of available denominations
61.08	Hardware	Withdraw when ATM has less cash than requested	Transaction declined; no partial dispense; account not debited
61.09	Cash not taken	Take card but do not collect dispensed cash	Cash retracted after timeout; transaction recorded per policy
61.10	Timeout	Leave machine idle on PIN screen	Session ends; card returned; no access to account
61.11	Receipt	Choose receipt print	Receipt shows correct amount, balance, date, machine id
61.12	Balance enquiry	Perform balance enquiry	Correct balance displayed; no cash dispensed
61.13	Exception	Power fails mid-transaction (before/after debit)	Either transaction completes or fully reverses; card not stuck
61.14	Exception	Block the card slot / shutter jams	Safe error state; support alert triggered

Edge Cases

Edge ID	Edge Case	Expected Behaviour
61-E1	Withdraw exactly full account balance	Dispenses if within limits; balance becomes ₹0 correctly
61-E2	Request more notes than the machine physically holds	Clean 'unavailable' message; no partial dispense
61-E3	Card pulled out early, mid-transaction	Transaction aborts safely; card not stuck; account consistent
61-E4	Power restored after mid-transaction failure	State recovered or transaction reversed; no lost money
61-E5	Withdraw daily limit exactly, then balance enquiry	Limit exhausted message is accurate; enquiry still works

62. How would you test a lift/elevator?

Test Scenarios

- Door operations (open, close, obstruction)
- Floor selection and correct travel
- Overload / capacity alarm
- Emergency and safety features (alarm, stop, power failure)
- Multi-lift coordination and edge conditions (top/bottom floors, idle)

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
62.01	Positive	Press a floor button inside the lift	Lift moves only in the chosen direction to that floor
62.02	Doors	Press Open while doors are closing	Doors reopen
62.03	Doors	Stand in the door path while closing	Doors detect obstruction and reopen; no injury/pressure
62.04	Multiple floors	Press 3 floors in sequence (e.g., 2, 5, 3) while going up	Floors served in correct order: 2, 3, 5
62.05	Calling	Press Up call on floor 3 while lift is below and moving down	Lift completes current direction first, then serves floor 3
62.06	Overload	Load beyond rated capacity (e.g., +1 person over limit)	Alarm sounds; doors stay open; lift does not move until reduced
62.07	Capacity edge	Load exactly at rated capacity	Lift operates normally, no alarm
62.08	Emergency	Press emergency alarm/intercom	Alarm audible; intercom connects to security
62.09	Emergency	Press emergency stop between floors	Lift stops safely; stop releases and resumes correctly
62.10	Power failure	Cut power while moving	Brakes engage; lift stops safely; backup light/intercom on
62.11	Boundaries	Press the topmost and bottom-most floors	Travels correctly; does not overshoot; sensors stop precisely
62.12	Wrong input	Press current floor's button	No movement
62.13	Idle	No calls for long time	Lift parks at default floor (e.g., ground) as per policy
62.14	Multi-lift	Two lifts idle; one call pressed	Nearest lift responds; not both

Edge Cases

Edge ID	Edge Case	Expected Behaviour
62-E1	Load exactly at rated capacity	No alarm; normal operation
62-E2	Overweight by a tiny margin (sensor tolerance)	Alarm triggers reliably within spec tolerance; no false passes
62-E3	Up and Down call pressed on the same floor simultaneously	Lift responds per algorithm; no deadlock
62-E4	Door-open pressed while lift is moving between floors	Ignored while moving; honoured only at floor level
62-E5	All floor buttons pressed, then one cancelled mid-travel	Remaining stops still served correctly
62-E6	Power returns while lift is stopped between floors	Recovers to nearest floor with doors opening; passengers exit safely

63. How would you test a calculator?

Test Scenarios

- Basic operations (add, subtract, multiply, divide)
- Decimal, negative and very large/small numbers
- Error handling (divide by zero, overflow, invalid input)
- Repeated/continuous operations and order of operations
- Memory, percentage, clear functions
- UI: keyboard support, display limits, history

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
63.01	Positive	2 + 3 = 5; 10 − 4 = 6; 6 × 7 = 42; 20 ÷ 4 = 5	Each result correct
63.02	Positive	Decimal: 1.5 + 2.25	3.75 displayed accurately
63.03	Negative numbers	5 − 10 = −5; −3 × 2 = −6	Signs handled correctly
63.04	Divide by zero	8 ÷ 0	'Error' or 'Cannot divide by zero' — no crash
63.05	Overflow	999999999 × 999999999	Overflow message or scientific notation, no silent wrong answer
63.06	Continuous	2 + 3 + 4 (press + 4 without = after first)	Running result 9 — chain operations correct
63.07	Repeated equals	5 = = = on '+' context	Behaviour matches design (repeat last operation)
63.08	Percent	200 + 10%	22 (percent applied to 200) as per design
63.09	Clear	Use C and CE mid-calculation	CE clears entry only; C clears all
63.10	Memory	M+ store, MR recall, MC clear	Stored value recalled and cleared correctly
63.11	UI	Type with keyboard (numbers, +, −, Enter)	Keys map correctly to operations
63.12	Display	Result with many decimals (e.g., 1 ÷ 3)	Rounded/displayed per design without layout break
63.13	Invalid input	Two operators in a row (5 + + 3)	Handled per design — ignored, error, or carried

Edge Cases

Edge ID	Edge Case	Expected Behaviour
63-E1	0 ÷ 0	Handled as error/undefined — never a number or crash
63-E2	(1 ÷ 3) × 3	Displays 1 (or consistent rounding), not 0.999…
63-E3	Result exactly at display digit limit	Scientific notation or clean truncation per design; no layout break
63-E4	Smallest minus largest value (underflow)	Handled with error/notation; no silent wraparound
63-E5	Percentage of a percentage (10% of 10%)	Computes per defined formula; result is 1, not 20

64. How would you test Google Search?

Test Scenarios

- Basic search with a simple valid query
- Empty, whitespace and special-character queries
- Autocomplete/suggestions while typing
- Results relevance, count and pagination
- Advanced features: operators, filters, tabs (Images, News), voice
- Performance, localization and error-resilience

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
64.01	Positive	Search a common keyword (e.g., 'weather')	Relevant results returned quickly with headers/snippets
64.02	Empty	Search empty string or only spaces	No search performed or handled gracefully
64.03	Special chars	Search with symbols @#$% and long strings	Handled without error; no crash
64.04	Autocomplete	Type partial word and observe suggestions	Relevant suggestions appear; selectable; search runs on select
64.05	Relevance	Search an exact phrase in quotes	Top results contain the exact phrase
64.06	No results	Search a nonsense/gibberish string	'No results found' page with helpful suggestions
64.07	Filters	Filter results by time (past year) and type	Only results matching the filter shown
64.08	Tabs	Switch to Images / News / Videos tabs	Tab-specific results load correctly
64.09	Pagination	Go to page 2, 3 and use navigation	Next set of results; no duplication of earlier results
64.10	Voice	Use voice search	Speech captured and results for spoken query shown
64.11	Performance	Measure response time of a search	Results render within the expected time (e.g., <2s typical)
64.12	Localization	Search in a regional language / from another locale	Results and UI adapt to language/region
64.13	Autocorrect	Search a misspelled word (e.g., 'googel')	'Did you mean' correction offered with better results
64.14	Site operator	Search 'site:example.com keyword'	Only results from that domain returned

Edge Cases

Edge ID	Edge Case	Expected Behaviour
64-E1	Query made only of stop words ('the', 'a')	Returns results or a sensible 'no results' — no error
64-E2	Extremely long query (32+ words)	Handled per limit; no crash or truncated request error
64-E3	Query mixing scripts (English + regional language)	Results from both languages as designed
64-E4	Autocomplete suggestions for special characters mid-typing	Suggestions behave safely; no injection into the query
64-E5	Search exactly while backend is slow/timing out	Retry or graceful failure message; page never hangs silently

65. How would you test a mobile application?

Test Scenarios

- Installation, launch, update and uninstall
- Functional testing of core features on real devices
- Device/OS fragmentation (screen sizes, OS versions)
- Interruption testing (calls, SMS, notifications, low battery, network loss)
- Non-functional: performance, battery, data usage, memory
- Permissions, offline mode and notifications

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
65.01	Install	Install app fresh on Android and iOS	Installs without error; icon appears; opens correctly
65.02	Update	Update from an older version keeping data	Data preserved; new version works; no crash on first launch
65.03	Uninstall	Uninstall and reinstall	Clean state; login works again
65.04	Fragmentation	Run on small (4.7") and large (6.7") screens, min and latest OS	UI fits, no cut-off text, no overlapping elements
65.05	Interruption	Receive a call / SMS while using the app	App suspends gracefully; resumes in same state after call
65.06	Interruption	Put app in background during a transaction, restore it	State retained or safely restarted; no data corruption
65.07	Network	Turn off data mid-action (e.g., during submit)	Friendly offline message; no crash; retries when back online
65.08	Battery	Run app 30 min; check battery and heat	Drain and temperature within acceptable limits
65.09	Performance	Measure launch time, scroll smoothness, memory (leaks)	Meets targets; no ANR/OOM; stable over long session
65.10	Permissions	Deny camera/location permission when prompted	Feature degrades gracefully; explains need; can be granted later
65.11	Offline	Open app in airplane mode	Cached content or clear offline message as designed
65.12	Notifications	Trigger a push notification (app open, background, killed)	Notification arrives and deep-links to the right screen
65.13	Orientation	Rotate to landscape / back	Layout adapts or locks as designed; no state loss
65.14	Stability	Monkey/long-run usage across features	No crashes; crash-free session rate within target

Edge Cases

Edge ID	Edge Case	Expected Behaviour
65-E1	Incoming call during a payment transaction	App suspends and resumes without duplicating or losing the transaction
65-E2	OS update installed while app is in background	App works correctly on the new OS version
65-E3	Time zone changed while app is open	Times/timestamps re-render correctly; no crash
65-E4	Device storage full during a file save/download	Clear 'storage full' message; no silent data loss
65-E5	Locale switched to a right-to-left (RTL) language	Layout mirrors correctly; text readable, nothing overlapping
65-E6	App killed by OS (low memory) mid-flow and reopened	Restores or safely restarts state; no data corruption

66. How would you test a logout functionality?

Test Scenarios

- Logout from an active session
- Session invalidation after logout (back button, URL access, API)
- Logout across multiple tabs/devices
- Logout during in-flight activity (transaction, upload)
- Auto-logout on inactivity timeout

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
66.01	Positive	Click Logout while logged in	User logged out; returned to login page; name hidden
66.02	Session	Press browser Back after logout	Login page shown again; no access to authenticated pages
66.03	Session	Paste an inner (authenticated) URL after logout	Redirected to login; access denied
66.04	API	Reuse the old auth token in an API call after logout	401 Unauthorized — token invalidated on server
66.05	Tabs	Logout in one tab while app is open in another	Other tab also logged out on next action (or per policy)
66.06	Multi-device	Logout from mobile while logged in on web	Web behaviour per spec (all sessions or per-device)
66.07	In-flight	Logout while a file upload/payment is in progress	Handled cleanly — cancelled or completed, no data corruption
66.08	Inactivity	Leave the app idle beyond timeout (e.g., 15 min)	Auto-logout; next action prompts login
66.09	Data	Check cart / unsaved data behaviour after logout	Data persists for the account or cleared as per spec
66.10	Repeat	Click Logout twice or when session already expired	No error; consistent login-page state

Edge Cases

Edge ID	Edge Case	Expected Behaviour
66-E1	Logout clicked while an API request is in flight	Request either completes or is cancelled cleanly; no orphan session
66-E2	Logout from two tabs at the same moment	Both end in logged-out state; no stuck session or error
66-E3	Auto-logout timer expires while the user is actively typing	Activity resets timer — user is not logged out mid-use
66-E4	Logout attempted while offline	Queued/local logout or clear message; consistent state on reconnect
66-E5	Token refresh happens exactly during logout	No resurrected session; refresh token also invalidated

67. How would you test a password change functionality?

Test Scenarios

- Successful change with correct current password
- Wrong current password rejection
- New password policy enforcement
- Confirm-password mismatch
- Old/new password rules and re-login behaviour
- Security notifications and other-session invalidation

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
67.01	Positive	Enter correct current password + valid new password + matching confirm	Password changed; confirmation message
67.02	Negative	Enter wrong current password	'Current password incorrect'; change blocked
67.03	Policy	New password violating rules (short, no digit/special)	All rule failures listed
67.04	Policy	New password exceeds max length or is same as old	Rejected per policy with clear message
67.05	Mismatch	Confirm password differs from new password	'Passwords do not match' error
67.06	Empty	Submit with one or more fields empty	Field-level required messages
67.07	Old invalid	Try logging in with old password after change	Login fails
67.08	New valid	Log in with the new password	Login succeeds
67.09	Sessions	After change, use other logged-in devices	Sessions invalidated or forced re-login as per policy
67.10	Notification	Check email/SMS after change	Alert received; no password written in plaintext in it
67.11	Masking	Verify all three fields mask input	Dots shown; show/hide toggles work per field
67.12	Rate limit	Attempt many rapid changes	Throttled per policy

Edge Cases

Edge ID	Edge Case	Expected Behaviour
67-E1	New password exactly equal to current password	Rejected per policy with specific message
67-E2	New password exactly at min / max length boundary	min accepted, min−1 rejected; max accepted, max+1 rejected
67-E3	Unicode / special characters (emoji, non-ASCII) in new password	Accepted and stored exactly if policy allows; re-login works
67-E4	Change made on device A while device B has an active session	B invalidated or prompted to re-authenticate per policy
67-E5	Password change immediately followed by login with the old password	Old password is already invalid — no caching window

68. How would you test pagination?

Test Scenarios

- Correct page navigation (Next, Previous, page numbers, first/last)
- Correct items-per-page and total page count
- Data integrity across pages (no duplicates/missing records)
- Pagination combined with search, sort and filters
- Edge cases: single page, last page, invalid page number

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
68.01	Positive	Click Next on page 1	Page 2 loads with the next set of records
68.02	Positive	Click Previous on page 2	Returns to page 1 with the same records
68.03	Navigation	Jump to a specific page number and to Last/First	Correct page loads each time
68.04	Count	Verify total records: 100 items, 10 per page = 10 pages	Page count and item counts correct
68.05	Integrity	Walk pages 1→N collecting records	No record duplicated or missing across pages
68.06	Edge	Only one page of data exists	Pagination controls hidden or disabled
68.07	Edge	On last page, Next is disabled	No empty page beyond last
68.08	Edge	Manually enter URL with page=999 or page=-1 or page=abc	Graceful handling: redirected to valid page or 'no results'
68.09	Combination	Search a term, then paginate results	Pagination applies only to filtered results
68.10	Combination	Sort by price, go to page 2	Sort order maintained across pages
68.11	State	Change per-page size (10 → 50) on page 3	Resets to a valid page (page 1) with the new size
68.12	UI	Verify current-page highlight and previous/next states	Active page indicated; buttons disabled at ends

Edge Cases

Edge ID	Edge Case	Expected Behaviour
68-E1	Records deleted while the user is paginating	No empty/error page; user lands on a valid page with correct data
68-E2	Last page contains exactly one record; it is deleted	User redirected to the previous page, not a blank page
68-E3	Page size set equal to total record count	Single-page view; navigation hidden/disabled cleanly
68-E4	Page number 0, negative, or non-numeric in URL	Graceful redirect or 'not found' — never a crash or empty grid
68-E5	Sort order changed while sitting on the last page	User moved to a still-valid page with the new order applied

69. How would you test sorting?

Test Scenarios

- Each sort option (price low-high/high-low, name A-Z/Z-A, newest)
- Numeric vs lexicographic correctness (2 vs 10 pitfall)
- Sort stability and combined with pagination
- Mixed/edge data (nulls, duplicates, symbols, long text)
- Sort indicators and toggling

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
69.01	Positive	Sort by price low → high	List ordered ascending; each price ≤ next
69.02	Positive	Sort by price high → low	Descending order correct
69.03	Positive	Sort by name A → Z and Z → A	Alphabetical order correct both ways
69.04	Positive	Sort by newest first	Latest date/time items on top
69.05	Numeric pitfall	Items priced 2 and 10 — check order	2 before 10 (numeric), NOT 10 before 2 (text sort)
69.06	Pagination	Sort then go to page 2	Order continues correctly across pages
69.07	Toggle	Click the same column twice	Order flips ascending ↔ descending; arrow indicator updates
69.08	Mixed data	Sort with null/empty values, same values, symbols	Deterministic placement per spec (end/top of list)
69.09	Duplicates	Many rows with equal keys	Stable or defined secondary order; no flicker
69.10	Combined	Filter category, then sort by price	Sorting applies within filtered results
69.11	Large data	Sort a large list (e.g., 10k rows)	Completes in acceptable time; correct order
69.12	Default	Load page without sorting	Default order per spec (e.g., relevance/popularity)

Edge Cases

Edge ID	Edge Case	Expected Behaviour
69-E1	All rows have identical sort keys	Stable, deterministic order; no flicker or reshuffling between loads
69-E2	Mixed data in a numeric column (some values blank/text)	Blanks and text placed per spec (end or top) — consistently
69-E3	Negative values (prices, dates before 1970, negative ratings)	Sorted correctly relative to zero and positives
69-E4	Unicode / special-character names (É, ü, emoji)	Sorted per locale collation or ASCII consistently; no crash
69-E5	Sort applied at the same moment data updates	No torn results; either old or new consistent ordering shown

70. How would you test filtering?

Test Scenarios

- Single filter application (category, price range, brand)
- Multiple filters combined (AND behaviour)
- Clearing filters and default 'All' state
- Filter + search / sort / pagination combinations
- No-results and invalid-range handling
- Result count accuracy and persistence

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
70.01	Positive	Apply one filter (category = Mobiles)	Only mobiles listed
70.02	Positive	Apply price range ₹10,000–₹20,000	All results within range, inclusive of both ends
70.03	Combined	Category = Mobiles AND Brand = Samsung AND price range	All three conditions satisfied by every result (AND logic)
70.04	Count	Compare result count with filter ON vs filter label count	Counts match and update on every change
70.05	Clear	Click Clear Filters / Reset	All products shown again; filter controls reset
70.06	No results	Apply a filter combination with zero matches	'No results found' with helpful message; option to clear filters
70.07	Invalid	Enter min price > max price	Blocked or auto-corrected with a message; no broken grid
70.08	Combination	Filter + search keyword	Both applied together
70.09	Combination	Filter, then paginate	Filter persists across pages; no unfiltered items leak in
70.10	Combination	Filter, then sort by price	Sorting within the filtered set
70.11	State	Refresh page / return via Back	Filter state retained or reset consistently per design
70.12	Multi-select	Select multiple values in one filter (brands A + B)	Union of both brands shown
70.13	Performance	Apply many filters on a large catalogue	Response within acceptable time

Edge Cases

Edge ID	Edge Case	Expected Behaviour
70-E1	Product priced exactly at the filter boundary (₹10,000)	Included when boundary is inclusive — tested on both ends
70-E2	Min price = max price in the range filter	Returns exactly the items at that price
70-E3	Filter that matches exactly one product	Single result shown; grid and count consistent
70-E4	All filters cleared at once vs one at a time	Same final result set either way
70-E5	Rapid filter clicks before responses return	UI shows the latest selection's results; no race/stale grid
70-E6	Filters applied while new products are being added	Results reflect one consistent snapshot

71. How would you test a notification feature?

Test Scenarios

- Notification triggered by the right event
- Delivery and timing (real-time, background, killed app)
- Content correctness (title, message, image, deep link)
- User actions: tap, dismiss, badge count, clear-all
- Settings and permissions (sound, vibration, opt-out, DND)

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
71.01	Positive	Perform the trigger event (e.g., order placed)	Notification received on time with right title and message
71.02	Content	Verify notification details (title, body, image, timestamp)	Content correct; no wrong or missing data
71.03	Deep link	Tap the notification	App opens directly to the relevant screen (order details)
71.04	States	Trigger with app in foreground vs background vs killed	Received/handled in all three states per design
71.05	Badge	Receive 3 notifications without opening	Badge count = 3; opens/clears reset it to 0
71.06	Dismiss	Swipe away / clear all	Notification removed; no ghost entries
71.07	Permission	Deny notification permission, then trigger	No notification shown; in-app fallback per design
71.08	Settings	Disable notifications in app settings	No push received; re-enabling restores them
71.09	Sound/vibration	Trigger with sound ON and OFF settings	Behaviour follows settings
71.10	DND	Trigger while device is in Do Not Disturb	Respected per OS rules; critical alerts per policy
71.11	Duplicates	Trigger the event once	Exactly one notification — no duplicates
71.12	Timing	Check delivery delay after event	Within expected SLA (near real-time)
71.13	Special chars	Notification text with emojis / long text / special characters	Renders correctly; truncated elegantly

Edge Cases

Edge ID	Edge Case	Expected Behaviour
71-E1	Notification arrives while user is on the notification-settings screen	Delivered/updated normally; no blocking or loop
71-E2	Notification during an ongoing phone call	Sound suppressed or shown per OS rules; not lost
71-E3	100+ unread notifications — badge count display	Shows '99+' or exact count per design; layout intact
71-E4	Notification when device storage is full	Not silently lost; retried or in-app fallback
71-E5	Two events triggered in the same second	Both delivered or aggregated per design; no dropped notification
71-E6	Event at midnight crossing time zones	Timestamp shown in the user's current zone correctly

72. How would you test a shopping coupon/discount?

Test Scenarios

- Valid coupon applies correctly
- Invalid / expired / wrong-case coupon rejection
- Terms enforcement (min cart value, specific products, one-time use)
- Discount maths (percentage, flat, capped) and totals
- Interaction with cart changes, stacking and other discounts

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
72.01	Positive	Apply valid coupon SAVE10 (10% off)	10% deducted; totals recalculated correctly
72.02	Positive	Apply flat coupon ₹200 off	₹200 deducted; grand total correct
72.03	Negative	Enter invalid code SAVE99	'Invalid coupon' error; no discount applied
72.04	Negative	Apply expired coupon	'Coupon expired' message
72.05	Case	Enter valid code in lowercase 'save10'	Applied (case-insensitive) or clear message per spec
72.06	Min value	Coupon needs ₹500 min cart; cart is ₹400	Rejected with 'minimum cart value ₹500' message
72.07	Scope	Coupon valid only on Fashion; cart has Electronics	Rejected or applied only to fashion items per design
72.08	One-time	Reuse a per-user coupon that was already used	'Coupon already used' error
72.09	Cap	50% off coupon capped at ₹1,000 on a ₹10,000 cart	Discount exactly ₹1,000, not ₹5,000
72.10	Stacking	Try two coupons together	Blocked (or combined per policy) — consistent
72.11	Cart change	Apply coupon, then remove the qualifying item	Coupon auto-removed or flagged; totals updated
72.12	Maths	Verify order total = subtotal − discount + tax/shipping	Arithmetic exactly correct; rounding handled
72.13	Empty	Apply blank coupon code	Validation error; no server call
72.14	Refund	Return an item bought with a coupon	Refund adjusted per coupon policy

Edge Cases

Edge ID	Edge Case	Expected Behaviour
72-E1	Coupon expires exactly while user is on the payment step	Consistent outcome: honoured or rejected with clear message — never random
72-E2	Discount greater than item price (e.g., flat ₹300 off a ₹250 item)	Total clamped at ₹0; no negative total; rest not discounted
72-E3	Cart exactly at the minimum-value boundary (₹500)	Coupon accepted only if boundary is inclusive — verified both sides
72-E4	10% off on ₹99.99 (rounding edge)	Rounds to ₹99.99 or ₹100.00 strictly per spec; totals always consistent
72-E5	Coupon applied, then all qualifying items removed	Coupon auto-removed or flagged; no invalid discount at payment
72-E6	Coupon on an already 100%-discounted (free) item	No negative price; coupon ignored or rejected per policy

73. How would you test a product checkout flow?

Test Scenarios

- Happy path: cart → address → payment → confirmation
- Guest vs logged-in checkout
- Address and delivery-option handling
- Order summary accuracy (items, totals, tax, shipping)
- Failure paths: payment failure, empty cart, stock change
- Post-order: confirmation, history, notification

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
73.01	Happy path	Add item → checkout → address → pay → confirm	Order placed; order id generated; success page
73.02	Guest	Checkout as guest vs logged-in user	Both supported; guest asked to log in/register per design at right step
73.03	Address	Add / edit / select saved address, including pincode validation	Address saved and shown in summary; serviceable pincodes accepted
73.04	Delivery	Select delivery speed options (standard/express)	Correct shipping fee and ETA reflected in summary
73.05	Summary	Verify items, qty, price, discount, tax, shipping, grand total	Every figure matches the cart and rules
73.06	Empty cart	Open checkout URL with empty cart	Blocked with 'cart is empty' redirect
73.07	Stock	Item goes out of stock before payment	Checkout blocked with clear message; no payment taken
73.08	Payment fail	Use a failing payment method	Order not confirmed; cart preserved; retry allowed
73.09	Edit at checkout	Change quantity / remove item on the checkout page	Totals update; flow continues consistently
73.10	Confirmation	Check order confirmation screen, email and SMS	All show correct order id, items and amount
73.11	History	Open My Orders after checkout	New order appears with correct status
73.12	Interruption	Refresh / lose connection mid-checkout	Session resumes or fails safely; no duplicate order
73.13	Duplicate	Double-click Place Order	Exactly one order created; no double payment

Edge Cases

Edge ID	Edge Case	Expected Behaviour
73-E1	Last unit of stock: two users checkout simultaneously	One order succeeds; other blocked with 'out of stock' — never oversold
73-E2	Payment succeeds but confirmation page fails to load	Order exists with correct status; visible in My Orders; no duplicate
73-E3	Non-serviceable / remote pincode entered at address step	Blocked with clear message and option to change address
73-E4	All items removed at the payment step	Checkout aborted cleanly; no empty order created
73-E5	Order placed exactly at midnight (delivery date boundary)	Delivery date belongs to exactly one day's cutoff logic
73-E6	Session times out while entering the address	Data preserved or cleanly re-entered; no broken half-order

74. How would you test an API response manually?

Test Scenarios

- Send request and validate status code + response body
- Validate response headers and structure (schema, field types)
- Positive and negative input handling
- Auth, method and not-found errors (401/405/404)
- Performance (response time), data consistency and idempotency

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
74.01	Positive	Send GET with valid params in Postman	200 OK; body matches expected data and correct JSON structure
74.02	Validation	Check every field's name, type and value in the response	No missing/extra fields; types match the API contract
74.03	Negative	Send invalid input values (bad format, out-of-range)	400 with a clear, structured error message and field details
74.04	Mandatory	Omit a required field	400 with 'field is required'; no partial data created
74.05	Auth	Call without token / with expired token	401 Unauthorized with proper error body
74.06	Method	Call GET endpoint with POST (or wrong verb)	405 Method Not Allowed
74.07	Not found	Request a non-existent resource id	404 with clean error message; no stack trace leaked
74.08	Headers	Verify Content-Type, Cache-Control, etc.	Headers correct per contract
74.09	Timing	Measure response time for normal load	Within agreed SLA (e.g., <1–2s)
74.10	DB check	Compare response data with database values	API data matches stored data exactly
74.11	Idempotency	Submit the same request twice (with idempotency key where applicable)	No duplicate resource created
74.12	Boundary	Send boundary values (0, max lengths, empty list)	Handled per contract — accepted or rejected consistently
74.13	Security	Try SQL/script payloads in parameters	Sanitized; no injection; no sensitive data in response

Edge Cases

Edge ID	Edge Case	Expected Behaviour
74-E1	Response arrives exactly at the timeout boundary	Classified consistently as success or timeout; no partial handling
74-E2	Payload with Unicode / emojis / special characters	Stored and returned exactly as sent (UTF-8 intact)
74-E3	Field that is an empty array vs null vs missing	Each case handled per contract — distinguished correctly
74-E4	Leap-day date (29/02) in a date field	Validated correctly against leap and non-leap years
74-E5	Rapid requests beyond the rate limit	429 returned with retry-after info; no 500 errors
74-E6	Two identical concurrent PUT requests (race)	No duplicate/corrupted record; one defined final state

75. How would you test a download button?

Test Scenarios

- Successful download of the correct file
- File integrity (name, format, size, content opens)
- Cancel, failure and no-network handling
- Repeated / concurrent downloads and large files
- Browser/permission behaviour and progress indication

Test Cases

TC ID	Scenario	Test Case Description	Expected Result
75.01	Positive	Click download on a small file	File downloads successfully to default folder
75.02	Correctness	Verify filename and extension (e.g., report.pdf not .tmp)	Exact expected name and format
75.03	Integrity	Open the downloaded file	Opens without corruption; content matches the source
75.04	Size	Compare file size with the source file	Byte size matches
75.05	Cancel	Click Cancel mid-download	Stops cleanly; partial file removed or marked .crdownload
75.06	No network	Disconnect network and click download	Clear error message; no corrupt file saved
75.07	Failure	Trigger server error (500) on the file link	Friendly error; no empty/corrupt file downloaded
75.08	Large file	Download a very large file (e.g., 500 MB)	Completes; progress % and speed shown; no browser freeze
75.09	Repeated	Click the button twice quickly	One file (or '(1)' suffix) — no corrupted interleaved file
75.10	Concurrent	Download two different files simultaneously	Both complete correctly
75.11	UI	Verify button states: hover, disabled while downloading, spinner	States change appropriately
75.12	Permissions	Download with browser location permission / 'ask each time' set	Browser prompt appears; works after allowing
75.13	Compat	Test on Chrome, Firefox, Edge and mobile	Downloads work on all supported browsers

Edge Cases

Edge ID	Edge Case	Expected Behaviour
75-E1	Network drops at ~99% of a large download	Partial file cleaned or resumable; clear failure message
75-E2	Device storage full mid-download	Clear error; partial file removed or marked incomplete
75-E3	Filename contains special characters, spaces, or emojis	Sanitized per OS rules; file still opens correctly
75-E4	Download the same filename twice	Second saved with '(1)' suffix; first not overwritten
75-E5	Server returns a 0-byte file with 200 OK	Detected and reported as a failed/empty download
75-E6	Download clicked while network permission dialog is pending	Resumes correctly after permission granted; no stuck state